2025-11-14 - Servidio (13)

Previous lecture: 2025-11-07 - Valentini (12)
Next lecture: 2025-11-27 - Valentini (14)

Quadratic invariance of incompressible ideal MHD

Quadratic form of MHD invariance
(Biskmap book)

First invariant

Energy

$$E \equiv \left\langle \frac{v^2 + b^2}{2} \right\rangle = \frac{1}{V} \int d^3x \left[\frac{U^2(\bar{x}, t)}{2} + \frac{B^2(\bar{x}, t)}{2} \right]$$

which has an apparent time dependency.

Second invariant

Cross helicity

$$H_c \equiv \left\langle \frac{\bar{U} \cdot \bar{B}}{2} \right\rangle$$

Third invariant

Magnetic helicity

$$H_m \equiv \left\langle \frac{\bar{a} \cdot \bar{b}}{2} \right\rangle$$

with $\bar{a}$ as the magnetic potential

$$\nabla \times \bar{a} = \bar{b}$$

Usually $\bar{b}$ is the fluctuation of magnetic field, while $\bar{B}$ is the full magnetic field.
We use the following hypotesys

$$\langle \bar{U} \rangle = \langle \bar{b} \rangle = 0$$

MHD is galileian invariant.

Conservation of energy

$$\frac{d}{dt} E(t) = 0$$

$$
\begin{aligned}
\frac{d}{dt}E &= \frac{d}{dt}\left\langle \frac{U^2(\bar{x},t)}{2} + \frac{b^2(\bar{x},t)}{2} \right\rangle = \\
&= \frac{d}{dt}\left\{ \frac{1}{V}\int d^3x \left[\frac{U^2(\bar{x},t)}{2} + \frac{b^2(\bar{x},t)}{2} \right] \right\} = \\
&= \frac{1}{V}\int d^3x \frac{\partial}{\partial t}\left[\frac{U^2(\bar{x},t)}{2} + \frac{b^2(\bar{x},t)}{2} \right] \\
&= \left\langle \frac{1}{2}\frac{\partial}{\partial t}\left[U^2(\bar{x},t) + b^2(\bar{x},t) \right] \right\rangle = \\
&= \left\langle \bar{U}\cdot\frac{\partial \bar{U}}{\partial t} + \bar{b}\cdot\frac{\partial \bar{b}}{\partial t} \right\rangle
\end{aligned}
$$

We use incompressible MHD equations.
HERE why we sub $\bar{B}$ with $\bar{b}$ without any correction? or why $(\nabla\times\bar{b})\times\bar{b} = (\nabla\times\bar{B})\times\bar{B}$?

$$
\begin{cases}
\dfrac{\partial \bar{U}}{\partial t} = -\bar{U}\cdot\nabla\bar{U} - \nabla P + (\nabla\times\bar{b})\times\bar{b} \\[2ex]
\dfrac{\partial \bar{b}}{\partial t} = \nabla\times(\bar{U}\times\bar{b}) = -\bar{U}\cdot\nabla\bar{B} + \bar{B}\cdot\nabla\bar{U}
\end{cases}
$$

We substitute

$$
\begin{aligned}
\bar{U}\cdot\frac{\partial \bar{U}}{\partial t} &= \bar{U}\cdot\left[-\bar{U}\cdot\nabla\bar{U} + (\nabla\times\bar{b})\times\bar{b} - \nabla P\right] = \\
&= -\bar{U}\cdot[\bar{U}\cdot\nabla\bar{U}] + \bar{U}\cdot[(\nabla\times\bar{b})\times\bar{b}] - \bar{U}\cdot\nabla P
\end{aligned}
$$

We have incompressibility, so

$$
\nabla\cdot\bar{U} = 0 \qquad \Longrightarrow \qquad \bar{U}\cdot\nabla\bar{U} = \nabla\cdot(\bar{U}\bar{U})
$$

(we can move $\bar{U}$ in divergence without any correction term)
and

$$
\bar{U}\cdot[\bar{U}\cdot\nabla\bar{U}] = \bar{U}\cdot[\nabla\cdot(\bar{U}\bar{U})] = \nabla\cdot(U^2\bar{U})
$$

and also

$$
\begin{aligned}
\bar{U}\cdot[(\nabla\times\bar{b})\times\bar{b}] &= \bar{U}\cdot\left[\bar{b}\cdot\nabla\bar{b} - \nabla\left(\frac{b^2}{2}\right)\right] = \\
&= \bar{U}\cdot\left[\nabla\cdot(\bar{b}\bar{b}) - \nabla\left(\frac{b^2}{2}\right)\right] = \\
&= \nabla\cdot[(\bar{U}\cdot\bar{b})\bar{b}] - \nabla\cdot\left(\frac{b^2}{2}\bar{U}\right) = \\
&= \nabla\cdot\left[(\bar{U}\cdot\bar{b})\bar{b} - \frac{b^2}{2}\bar{U}\right]
\end{aligned}
$$

and also

$$
\bar{U}\cdot\nabla P = \nabla\cdot(P\bar{U})
$$

So we have

$$\bar{U}\cdot\frac{\partial\bar{U}}{\partial t}=\nabla\cdot\left\{-U^2\bar{U}+(\bar{U}\cdot\bar{b})\bar{b}-\frac{b^2}{2}\bar{U}-P\bar{U}\right\}$$

We also have (using $\nabla\cdot\bar{U}=\nabla\cdot\bar{b}=0$)

$$\begin{aligned}\bar{b}\cdot\frac{\partial\bar{b}}{\partial t}&=\bar{b}\cdot\left[\nabla\times(\bar{U}\times\bar{b})\right]=\\&=\bar{b}\cdot\left[-\bar{U}\cdot\nabla\bar{b}+\bar{b}\cdot\nabla\bar{U}\right]=\\&=\bar{b}\cdot\left[-\nabla\cdot(\bar{U}\bar{b})+\nabla\cdot(\bar{b}\bar{U})\right]=\\&=\nabla\cdot\left[-(\bar{b}\cdot\bar{U})\bar{b}+b^2\bar{U}\right]=\\&=-\nabla\cdot\left[(\bar{b}\cdot\bar{U})\bar{b}-b^2\bar{U}\right]\end{aligned}$$

So

$$\begin{aligned}\bar{U}\cdot\frac{\partial\bar{U}}{\partial t}+\bar{b}\cdot\frac{\partial\bar{b}}{\partial t}&=\nabla\cdot\left\{-U^2\bar{U}+\cancel{(\bar{U}\cdot\bar{b})\bar{b}}-\frac{b^2}{2}\bar{U}-P\bar{U}-\cancel{(\bar{b}\cdot\bar{U})\bar{b}}+b^2\bar{U}\right\}=\\&=\nabla\cdot\left\{\bar{U}\left(\frac{b^2}{2}-U^2\right)-P\bar{U}\right\}\end{aligned}$$

we can use the divergence theorem

$$\frac{d}{dt}E=\frac{1}{V}\int\nabla\cdot\left\{\left(\frac{b^2}{2}-U^2\right)\bar{U}-P\bar{U}\right\}d^3x=\int\bar{M}\cdot\hat{n}\,d\Sigma$$

with

$$\bar{M}\equiv\left(\frac{b^2}{2}-U^2\right)\bar{U}-P\bar{U}$$

At boundaries $U_n\equiv\bar{U}\cdot\hat{n}=0$
And we have

$$\frac{d}{dt}E(t)=0\qquad\Longleftrightarrow\qquad U_n=0$$

Conservation of cross helicity

Cross helicity tells us how much $\bar{U}$ and $\bar{b}$ are aligned, averaged in space.
Recall

$$\begin{cases}\dfrac{\partial\bar{v}}{\partial t}=-\bar{U}\cdot\nabla\bar{U}-\nabla P+(\nabla\times\bar{b})\times\bar{b} & \text{(Euler equation)}\\[2ex]\dfrac{\partial\bar{b}}{\partial t}=\nabla\times(\bar{U}\times\bar{b})=-\bar{U}\cdot\nabla\bar{B}+\bar{B}\cdot\nabla\bar{U} & \text{(Ampère-Maxwell law)}\end{cases}$$

($\mathbb{1}$ is the identity tensor)

$$
\begin{aligned}
\frac{d}{dt}H_e = \frac{d}{dt}\left\langle \frac{\bar{U}\cdot\bar{b}}{2} \right\rangle &= \frac{1}{2}\left\langle \bar{b}\cdot\frac{\partial \bar{U}}{\partial t} + \bar{U}\cdot\frac{\partial \bar{b}}{\partial t} \right\rangle = \\
&= \frac{1}{2}\left\langle \bar{b}\cdot\nabla\cdot\left[-\bar{U}\bar{U} - P\mathbb{1} + \bar{b}\bar{b} - \frac{b^2}{2}\mathbb{1}\right] + \bar{U}\cdot\nabla\cdot\left[-\bar{U}\bar{b} + \bar{b}\bar{U}\right] \right\rangle = \\
&= \frac{1}{2}\left\langle \nabla\cdot\left[-\cancel{(\bar{U}\cdot\bar{b})\bar{U}} - P\bar{b} + b^2\bar{b} - \frac{b^2}{2}\bar{b} - U^2\bar{b} + \cancel{\bar{U}\cdot\bar{b}\bar{U}}\right] \right\rangle = \\
&= -\frac{1}{2}\left\langle \nabla\cdot\left[\bar{b}\left(P - \frac{b^2}{2} - U^2\right)\right] \right\rangle = \\
&= -\frac{1}{2}\left\langle \nabla\cdot\bar{A} \right\rangle
\end{aligned}
$$

with

$$
\bar{A} \equiv \bar{b}\left(P - \frac{b^2}{2} - U^2\right)
$$

so we have

$$
\frac{d}{dt}H_e = -\frac{1}{2V}\int d^3x\,\nabla\cdot\bar{A} = 0 \qquad \Longleftrightarrow \qquad b_n = 0
$$

minimal condition

Conservation of magnetic helicity

$$
\bar{a}: \quad \nabla\times\bar{a} = \bar{b}
$$

We have a gauge on $\bar{a}$

$$
\bar{a} = \bar{a}' + \nabla\phi
$$

with

$$
\bar{b} = \nabla\times\bar{a} \qquad \Longrightarrow \qquad \nabla\times\nabla\phi = 0
$$

and

$$
\bar{b} = \nabla\times\bar{a} = \nabla\times\bar{a}' \rightarrow \nabla\cdot\bar{a} = 0 \qquad \Longleftrightarrow \qquad \nabla\cdot\bar{a}' = -\nabla^2\phi
$$

So we evaluate

$$
\frac{d}{dt}H_m(t) = \frac{d}{dt}\left\langle \frac{\bar{a}\cdot\bar{b}}{2} \right\rangle = \frac{1}{2}\left\langle \frac{\partial(\bar{a}\cdot\bar{b})}{\partial t} \right\rangle = \frac{1}{2}\left\langle \bar{a}\frac{\partial \bar{b}}{\partial t} + \bar{b}\cdot\frac{\partial \bar{a}}{\partial t} \right\rangle
$$

We can write (using Ampère-Maxwell law)

$$
\frac{\partial \bar{b}}{\partial t} = -\nabla\times\bar{E} = \nabla\times(\bar{U}\times\bar{b})
$$

with

$$
\bar{E} = -\bar{U}\times\bar{b}
$$

and we also get

$$\frac{\partial \bar{b}}{\partial t} = \frac{\partial(\nabla \times \bar{a})}{\partial t} = \nabla \times \left(\frac{\partial \bar{a}}{\partial t}\right)$$

so we obtain

$$\frac{\partial \bar{a}}{\partial t} = \bar{U} \times \bar{b} + \nabla \psi$$

(ψ is any function differentiable and continuous in the considered volume).
So

$$\begin{aligned}\frac{\partial(\bar{a} \cdot \bar{b})}{\partial t} &= \bar{a} \cdot \frac{\partial \bar{b}}{\partial t} + \bar{b} \cdot \frac{\partial \bar{a}}{\partial t} = \\ &= -\bar{a} \cdot \left[-\bar{U} \cdot \nabla \bar{b} + \bar{b} \cdot \nabla \bar{U}\right] + \bar{b} \cdot \left[(\bar{U} \times \bar{b}) + \nabla \phi\right] = \\ &= \nabla \cdot \left[(\bar{a} \cdot \bar{b})\bar{U} - (\bar{a} \cdot \bar{U})\bar{b} + \phi \bar{b}\right]\end{aligned}$$

where the third term is null.

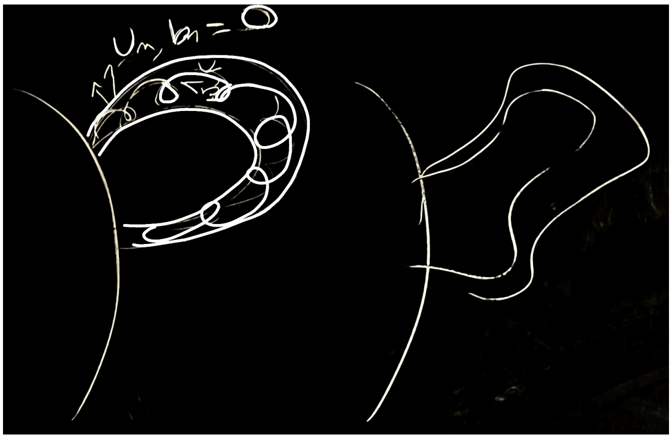
And finally

$$\frac{d}{dt} H_m = \frac{d}{dt}\left\langle \frac{\bar{a} \cdot \bar{b}}{2} \right\rangle = \frac{1}{2V} \int d^3x \, \nabla \cdot \left[(\bar{a} \cdot \bar{b})\bar{U} - (\bar{a} \cdot \bar{U})\bar{b} + \phi \bar{b}\right] = \frac{1}{2V} \int d^3x \, \nabla \cdot \bar{S}$$

$$\frac{d}{dt} H_m = 0 \qquad \Longleftrightarrow \qquad S_n = 0 \qquad \Longleftrightarrow \qquad \begin{cases} U_n = 0 \\ B_n = 0 \end{cases}$$

(also they balance out, but it's a very particular case)
Magnetic helicity measures twistedness, how much the magnetic field is twisting.

PIC sun, coronal mass ejection

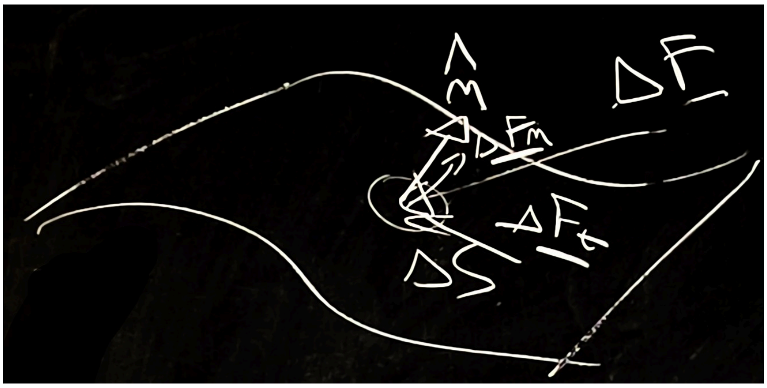


coronal loops, prominescence which exists for a long time
they obey these conservation laws
There are also forces that are not ideal, they convert energy from one source to others not considered in our model. They originate instabilities.
We introduce non-linear effects like viscosity and resistivity.

Momentum equation for non-ideal, incompresible MHD

We forget for a moment the magnetic field

PIC sheet of fluid



We have an infinitesiml surface of fluid, we decompose the internal forces acting on ΔS (the infinitesimal surface) in $F_{\parallel}$ to the surface and $F_{\perp}$ to the surface.

We have the following definitions for the pressure (P) and tagential stress (σ_t).

$$P = \lim_{\delta S \to 0} \frac{\Delta F_n}{\Delta S}; \qquad \sigma_t = \lim_{\Delta S \to 0} \frac{\Delta F_t}{\Delta S}$$

If $\nu = 0$ then $\sigma_t = 0$.

The stress tensor $\overleftrightarrow{\tau} = \tau_{ij}$ is

$$\tau_{ij} = -P\delta_{ij} + \frac{\nu}{2}\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right)$$

and (using $\nabla \cdot \overleftrightarrow{\tau} = \tau_{ij,i}\,\hat{e}_j$)

$$\begin{aligned}\frac{\partial \bar{U}}{\partial t} &= -\bar{U}\cdot\nabla\bar{U} + \nabla\cdot\overleftrightarrow{\tau} + (\nabla\times\bar{b})\times\bar{b} = \\ &= -\bar{U}\cdot\nabla\bar{U} - \nabla P + \nu\nabla^2\bar{U} + (\nabla\times\bar{b})\times\bar{b}\end{aligned}$$

where the term with the laplacian is the non-linear term.

Recall that

$$0 = \nabla\cdot\bar{U} = \frac{\partial U_i}{\partial x_i}; \qquad 0 = \nabla\left(\nabla\cdot\bar{U}\right) = \frac{\partial}{\partial x_j}\frac{\partial U_i}{\partial x_i}\hat{e}_j = \frac{\partial^2 U_i}{\partial x_j \partial x_i}\hat{e}_j$$

so

$$\begin{aligned}\nabla\cdot\overleftrightarrow{\tau} &= \tau_{ij,i}\,\hat{e}_j = \frac{\partial}{\partial x_i}\left\{-P\delta_{ij} + \frac{\nu}{2}\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right)\right\}\hat{e}_j = \\ &= -\frac{\partial}{\partial x_i}P\hat{e}_i + \cancel{\frac{\nu}{2}\frac{\partial^2 U_i}{\partial x_i \partial x_j}\hat{e}_j} + \frac{\nu}{2}\frac{\partial^2 U_j}{\partial x_i^2}\hat{e}_j = \\ &= -\nabla P + \nu\nabla^2\bar{U}\end{aligned}$$

where in the last step we used

$$\nabla P = \frac{\partial}{\partial x_i}P\hat{e}_i; \qquad \nabla^2\bar{U} = \frac{\partial^2}{\partial x_i^2}U_j\hat{e}_j$$

Induction equation for non-ideal MHD
We use Ohm's law in an Ohm's conductor tube

$$E\Delta x = \Delta V = Ri = R\bar{J} \cdot \bar{V} \qquad \Longrightarrow \qquad \bar{E} = \eta \bar{J}$$

with η resistivity
So the new, non ideal Ohm's law is

$$\bar{E} = -\bar{U} \times \bar{b} + \eta \bar{J}$$

from which

$$\begin{aligned}
\frac{\partial \bar{b}}{\partial t} &= -\nabla \times \bar{E} = \nabla \times \left[\bar{U} \times \bar{b} - \eta \bar{J}\right] = \\
&= \nabla \times (\bar{U} \times \bar{b}) - \eta \nabla \times (\nabla \times \bar{b}) = \\
&= \nabla \times (\bar{U} \times \bar{b}) - \eta \nabla^2 \bar{b} + \eta \nabla \cancel{(\nabla \cdot \bar{b})} = \\
&= \nabla \times (\bar{U} \times \bar{b}) - \eta \nabla^2 \bar{b}
\end{aligned}$$

So our non ideal, visco-resistive, incompressible MHD

$$\begin{cases}
\dfrac{\partial \bar{U}}{\partial t} = -\bar{U} \cdot \nabla \bar{U} - \nabla P + (\nabla \times \bar{b}) \times \bar{b} + \nu \nabla^2 \bar{U} \\
\dfrac{\partial \bar{b}}{\partial t} = \nabla \times (\bar{U} \times \bar{b}) + \eta \nabla^2 \bar{b}
\end{cases}$$

Previous lecture: 2025-11-07 - Valentini (12)
Next lecture: 2025-11-27 - Valentini (14)